

Experiment 2

Aim: To design and implement a relational database system using DBMS concepts by creating database schemas, defining tables and relationships, populating data, and executing SQL queries for efficient data retrieval and analysis.

1. - CREATE DATABASE stud1_match;
- USE stud1_match;
- CREATE TABLE ODIMatch (MatchID INT UNSIGNED, Team1 VARCHAR(15), Team2 VARCHAR(15), Ground VARCHAR(20), Date DATE, Winner VARCHAR(10), PRIMARY KEY (MatchID));
- DESCRIBE ODIMatch;

```
MySQL localhost:33060+ ssl SQL > CREATE DATABASE aarya61_match;
Query OK, 1 row affected (0.0123 sec)
MySQL localhost:33060+ ssl SQL > USE aarya61_match;
Default schema set to 'aarya61_match'.
Fetching global names, object names from 'aarya61_match' for auto-completion..
MySQL localhost:33060+ ssl aarya61_match SQL >
MySQL localhost:33060+ ssl aarya61_match SQL > CREATE TABLE ODIMatch (
-> MatchID INT UNSIGNED,
-> Team1 VARCHAR(15),
-> Team2 VARCHAR(15),
-> Ground VARCHAR(20),
-> Date DATE,
-> Winner VARCHAR(10),
-> PRIMARY KEY (MatchID)
-> );
Query OK, 0 rows affected (0.0424 sec)
MySQL localhost:33060+ ssl aarya61_match SQL > DESCRIBE ODIMatch;
```

Field	Type	Null	Key	Default	Extra
MatchID	int unsigned	NO	PRI	NULL	
Team1	varchar(15)	YES		NULL	
Team2	varchar(15)	YES		NULL	
Ground	varchar(20)	YES		NULL	
Date	date	YES		NULL	
Winner	varchar(10)	YES		NULL	

```
6 rows in set (0.0041 sec)
```

2. INSERT INTO ODIMatch (MatchID,Team1,Team2,Ground,Date,Winner) VALUES
 (2324,'Pakistan','India','Peshwar','2006-2-6','Team1'),
 (2327,'Pakistan','India','Rawalpindi','2006-2-11','Team2'),
 (2357,'India','England','Delhi','2006-3-28','Team1'),
 (2377,'West Indies','India','Kingston','2006-5-18','Team2'),
 (2404,'Sri Lanka','India','Colombo','2006-8-16','Abandoned'),

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(2440,'India','Australia','Mohali','2006-10-29','Team2'),
(2449,'South Africa','India','Cape Town','2006-11-26','Team1'),
(2480,'India','West Indies','Nagpur','2007-1-21','Team1'),
(2493,'India','west Indies','Vadodara','2007-1-31','Team1'),
(2520,'India','Sri Lanka','Rajkot','2007-2-11','Team2'),
(2611,'England','India','Southampton','2007-8-21','Team1'),
(2632,'India','Australia','Mumbai','2007-10-17','Team1'),
(2643,'India','Pakistan','Guwahati','2007-11-5','Team1'),
(2675,'Australia','India','Melbourne','2008-2-10','Team2'),
(2681,'India','Sri Lanka','Adelaide','2008-2-19','Team1'),
(2688,'Australia','India','Sydney','2008-3-2','Team2'),
(2689,'Australia','India','Brisbane','2008-3-4','Team2'),
(2717,'Pakistan','India','Karachi','2008-6-26','Team2'),
(2750,'Sri Lanka','India','Colombo','2008-8-24','Team2'),
(2755,'Sri Lanka','India','Colombo','2008-8-24','Team2');
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > INSERT INTO ODIMatch (MatchID,Team1,Team2,Ground,Date,Winner) VALUES
-> (2324,'Pakistan','India','Peshwar','2006-2-6','Team1'),
-> (2327,'Pakistan','India','Rawalpindi','2006-2-11','Team2'),
-> (2357,'India','England','Delhi','2006-3-28','Team1'),
-> (2377,'West Indies','India','Kingston','2006-5-18','Team2'),
-> (2404,'Sri Lanka','India','Colombo','2006-8-16','Abandoned'),
-> (2440,'India','Australia','Mohali','2006-10-29','Team2'),
-> (2449,'South Africa','India','Cape Town','2006-11-26','Team1'),
-> (2480,'India','West Indies','Nagpur','2007-1-21','Team1'),
-> (2493,'India','west Indies','Vadodara','2007-1-31','Team1'),
-> (2520,'India','Sri Lanka','Rajkot','2007-2-11','Team2'),
-> (2611,'England','India','Southampton','2007-8-21','Team1'),
-> (2632,'India','Australia','Mumbai','2007-10-17','Team1'),
-> (2643,'India','Pakistan','Guwahati','2007-11-5','Team1'),
-> (2675,'Australia','India','Melbourne','2008-2-10','Team2'),
-> (2681,'India','Sri Lanka','Adelaide','2008-2-19','Team1'),
-> (2688,'Australia','India','Sydney','2008-3-2','Team2'),
-> (2689,'Australia','India','Brisbane','2008-3-4','Team2'),
-> (2717,'Pakistan','India','Karachi','2008-6-26','Team2'),
-> (2750,'Sri Lanka','India','Colombo','2008-8-24','Team2'),
-> (2755,'Sri Lanka','India','Colombo','2008-8-24','Team2');

Query OK, 20 rows affected (0.0130 sec)

Records: 20 Duplicates: 0 Warnings: 0
```

3. SELECT * FROM ODIMatch;

MySQL localhost:33060+ ssl aarya61_match SQL > SELECT * FROM ODIMatch;

MatchID	Team1	Team2	Ground	Date	Winner
2324	Pakistan	India	Peshwar	2006-02-06	Team1
2327	Pakistan	India	Rawalpindi	2006-02-11	Team2
2357	India	England	Delhi	2006-03-28	Team1
2377	West Indies	India	Kingston	2006-05-18	Team2
2404	Sri Lanka	India	Colombo	2006-08-16	Abandoned
2440	India	Australia	Mohali	2006-10-29	Team2
2449	South Africa	India	Cape Town	2006-11-26	Team1
2480	India	West Indies	Nagpur	2007-01-21	Team1
2493	India	west Indies	Vadodara	2007-01-31	Team1
2520	India	Sri Lanka	Rajkot	2007-02-11	Team2
2611	England	India	Southampton	2007-08-21	Team1
2632	India	Australia	Mumbai	2007-10-17	Team1
2643	India	Pakistan	Guwahati	2007-11-05	Team1
2675	Australia	India	Melbourne	2008-02-10	Team2
2681	India	Sri Lanka	Adelaide	2008-02-19	Team1
2688	Australia	India	Sydney	2008-03-02	Team2
2689	Australia	India	Brisbane	2008-03-04	Team2
2717	Pakistan	India	Karachi	2008-06-26	Team2
2750	Sri Lanka	India	Colombo	2008-08-24	Team2
2755	Sri Lanka	India	Colombo	2008-08-24	Team2

20 rows in set (0.0010 sec)

4. CREATE TABLE Player (

PlayerID INT UNSIGNED,

LName VARCHAR(15),

FName VARCHAR(15),

Country VARCHAR(25),

Yborn INT,

Bplace VARCHAR(30),

Ftest INT,

PRIMARY KEY (PlayerID));

INSERT INTO Player (PlayerID,Lname,Fname,Country,Yborn,Bplace,Ftest)

VALUES (89001,'Tendulkar','Sachin','India',1973,'Mumbai',1989);

INSERT INTO Player (PlayerID,Lname,Fname,Country,Yborn,Bplace,Ftest)

VALUES (90001,'Lara','Brain','West Indies',1969,'Santa Cruz',1990);

INSERT INTO Player (PlayerID,Lname,Fname,Country,Yborn,Bplace,Ftest)

VALUES (95001,'Ponting','Ricky','Australia',1974,'Launceston',1995);

INSERT INTO Player (PlayerID,Lname,Fname,Country,Yborn,Bplace,Ftest)

VALUES (96001,'Dravid','Rahul','India',1973,'Indore',1996);

INSERT INTO Player VALUES

(96002,'Gibbs','Herschelle','South Africa',1974,'Cape Town',1996),

(92001,'Warne','Shane','Australia',1969,'Melbourne',1992),

(95002,'Pollock','Shaun','South Africa',1973,'Port Elizabeth',1995),

(99003,'Vaughan','Michael','England',1974,'Manchester',1999),

(92003,'UI-Huq','Inzamam','Pakistan',1970,'Multan',1992),

(94004,'Fleming','Stephen','New Zealand',1973,'Christchurch',1994),

(93002,'Streak','Heath','Zimbabwe',1974,'Bulawayo',1993),

(90002,'Kumble','Anil','India',1970,'Bangalore',1990),

(93003,'Kirsten','Gary','South Africa',1967,'Cape Town',1993),

(95003,'Kallis','Jacques','South Africa',1975,'Cape Town',1995),

(94002,'Vaas','Chaminda','Sri Lanka',1974,'Mattumagala',1994),

(92002,'Murlitharan','Muthiah','Sri Lanka',1972,'Kandy',1992),

(97004,'Vettori','Daniel','New Zealand',1979,'Auckland',1997),

(25001,'Dhoni','MS','India',1981,'Ranchi',2005),

(23001,'Singh','Yuvraj','India',1981,'Chandigarh',2003),

(96003,'Ganguly','Saurav','India',1972,'Calcutta',1996),

(99002,'Gilchrist','Adam','Australia',1971,'Bellingen',1999),

(24001,'Symonds','Andrew','Australia',1975,'Birmingham',2004),

(99001,'Lee','Breet','Australia',1976,'Wollongong',1999),

(91001,'Jayasuriya','Sanath','Sri Lanka',1969,'Matara',1991),

(21001,'Sehwag','Virender','India',1978,'Delhi',2001),

(98001,'Afridi','Shahid','Pakistan',1980,'Khyber Agency',1998),

(98002,'Singh','Harbhajan','India',1980,'Jalandhar',1998),

(27002,'Sharma','Ishant','India',1988,'Delhi',2007);

INSERT INTO Player (PlayerID,Lname,Fname,Country,Yborn,Bplace)

VALUES (27001,'Kumar','Praveen','India',1986,'Meerut');

```
MySQL localhost:33060+ ssl aarya61_match SQL > CREATE TABLE Player (
-> PlayerID INT UNSIGNED,
-> LName VARCHAR(15),
-> FName VARCHAR(15),
-> Country VARCHAR(25),
-> Yborn INT,
-> Bplace VARCHAR(30),
-> Ftest INT,
-> PRIMARY KEY (PlayerID));

Query OK, 0 rows affected (0.0225 sec)
MySQL localhost:33060+ ssl aarya61_match SQL >
MySQL localhost:33060+ ssl aarya61_match SQL > INSERT INTO Player (PlayerID,Lname,Fname,Country,Yborn,Bplace,Ftest)
-> VALUES (89001,'Tendulkar','Sachin','India',1973,'Mumbai',1989);

Query OK, 1 row affected (0.0037 sec)
MySQL localhost:33060+ ssl aarya61_match SQL >
MySQL localhost:33060+ ssl aarya61_match SQL > INSERT INTO Player (PlayerID,Lname,Fname,Country,Yborn,Bplace,Ftest)
-> VALUES (90001,'Lara','Brain','West Indies',1969,'Santa Cruz',1990);

Query OK, 1 row affected (0.0077 sec)
MySQL localhost:33060+ ssl aarya61_match SQL >
MySQL localhost:33060+ ssl aarya61_match SQL > INSERT INTO Player (PlayerID,Lname,Fname,Country,Yborn,Bplace,Ftest)
-> VALUES (95001,'Ponting','Ricky','Australia',1974,'Launceston',1995);

Query OK, 1 row affected (0.0027 sec)
MySQL localhost:33060+ ssl aarya61_match SQL >
MySQL localhost:33060+ ssl aarya61_match SQL > INSERT INTO Player (PlayerID,Lname,Fname,Country,Yborn,Bplace,Ftest)
-> VALUES (96001,'Dravid','Rahul','India',1973,'Indore',1996);

Query OK, 1 row affected (0.0022 sec)
MySQL localhost:33060+ ssl aarya61_match SQL >
MySQL localhost:33060+ ssl aarya61_match SQL > INSERT INTO Player VALUES
-> (96002,'Gibbs','Herschelle','South Africa',1974,'Cape Town',1996),
-> (92001,'Warne','Shane','Australia',1969,'Melbourne',1992),
-> (95002,'Pollock','Shaun','South Africa',1973,'Port Elizabeth',1995),
-> (99003,'Vaughan','Michael','England',1974,'Manchester',1999),
-> (92003,'UI-Huq','Inzamam','Pakistan',1970,'Multan',1992),
-> (94004,'Fleming','Stephen','New Zealand',1973,'Christchurch',1994),
-> (93002,'Streak','Heath','Zimbabwe',1974,'Bulawayo',1993),
-> (90002,'Kumble','Anil','India',1970,'Bangalore',1990),
-> (93003,'Kirsten','Gary','South Africa',1967,'Cape Town',1993),
-> (95003,'Kallis','Jacques','South Africa',1975,'Cape Town',1995),
-> (94002,'Vaas','Chaminda','Sri Lanka',1974,'Mattumagala',1994),
-> (92002,'Murlitharan','Muthiah','Sri Lanka',1972,'Kandy',1992),
-> (97004,'Vettori','Daniel','New Zealand',1979,'Auckland',1997),
-> (25001,'Dhoni','MS','India',1981,'Ranchi',2005),
-> (23001,'Singh','Yuvraj','India',1981,'Chandigarh',2003),
-> (96003,'Ganguly','Saurav','India',1972,'Calcutta',1996),
-> (99002,'Gilchrist','Adam','Australia',1971,'Bellingen',1999),
-> (24001,'Symonds','Andrew','Australia',1975,'Birmingham',2004),
-> (99001,'Lee','Breet','Australia',1976,'Wollongong',1999),
-> (91001,'Jayasuriya','Sanath','Sri Lanka',1969,'Matara',1991),
-> (21001,'Sehwag','Virender','India',1978,'Delhi',2001),
-> (98001,'Afridi','Shahid','Pakistan',1980,'Khyber Agency',1998),
-> (98002,'Singh','Harbhajan','India',1980,'Jalandhar',1998),
-> (27002,'Sharma','Ishant','India',1988,'Delhi',2007);

Query OK, 24 rows affected (0.0026 sec)
```

4. DESCRIBE Player;

```
MySQL localhost:33060+ ssl aarya61_match SQL > DESCRIBE Player;
```

Field	Type	Null	Key	Default	Extra
PlayerID	int unsigned	NO	PRI	NULL	
LName	varchar(15)	YES		NULL	
FName	varchar(15)	YES		NULL	
Country	varchar(25)	YES		NULL	
Yborn	int	YES		NULL	
Bplace	varchar(30)	YES		NULL	
FTest	int	YES		NULL	

7 rows in set (0.0042 sec)

5. CREATE TABLE Batting (
MatchID INT UNSIGNED,
PID INT UNSIGNED,
POrder TINYINT UNSIGNED,
HOut VARCHAR(5),
FOW VARCHAR(5),
NRuns SMALLINT UNSIGNED,

```

Mts SMALLINT UNSIGNED,
Nballs TINYINT UNSIGNED,
Fours TINYINT UNSIGNED,
Six TINYINT UNSIGNED,
FOREIGN KEY (MatchID) REFERENCES ODIMatch(MatchID),
FOREIGN KEY (PID) REFERENCES Player(PlayerID));

```

```

INSERT INTO Batting (MatchID,PID,POrder,Hout,FOW,NRuns,Mts,Nballs,Fours,Six)
VALUES (2755,23001,3,'C',51,0,12,6,0,0);

```

```

INSERT INTO Batting (MatchID,PID,POrder,Hout,FOW,NRuns,Mts,Nballs,Fours,Six)
VALUES (2755,25001,5,'C',232,71,104,80,4,0);

```

```

INSERT INTO Batting (MatchID,PID,POrder,Hout,FOW,NRuns,Mts,Nballs,Fours,Six)
VALUES (2755,91001,1,'C',74,60,85,52,8,2);

```

```

INSERT INTO Batting VALUES
(2755,94002,7,'LBW',157,17,23,29,1,0),
(2755,92002,11,'NO','NO',1,11,7,0,0),
(2689,89001,2,'C',205,91,176,121,7,0),
(2689,23001,4,'C',175,38,27,38,2,2),
(2689,25001,5,'C',240,36,52,37,2,1),
(2689,99002,1,'C',2,2,3,3,0,0),
(2689,95001,3,'C',8,1,9,7,0,0),
(2689,24001,5,'C',123,42,81,56,2,1),
(2689,99001,8,'B',228,7,20,12,0,0),
(2689,27001,9,'C',255,7,7,7,1,0),
(2755,27001,9,'B',257,2,7,6,0,0),
(2689,98002,8,'LBW',249,3,7,3,0,0),
(2755,98002,8,'RO',253,2,7,4,0,0);

```

```

MySQL localhost:33060+ ssl aarya61_match SQL > CREATE TABLE Batting (
--> MatchID INT UNSIGNED,
--> PID INT UNSIGNED,
--> POrder TINYINT UNSIGNED,
--> HOut VARCHAR(5),
--> FOW VARCHAR(5),
--> NRuns SMALLINT UNSIGNED,
--> Mts SMALLINT UNSIGNED,
--> Nballs TINYINT UNSIGNED,
--> Fours TINYINT UNSIGNED,
--> Six TINYINT UNSIGNED,
--> FOREIGN KEY (MatchID) REFERENCES ODIMatch(MatchID),
--> FOREIGN KEY (PID) REFERENCES Player(PlayerID));

Query OK, 0 rows affected (0.0392 sec)

MySQL localhost:33060+ ssl aarya61_match SQL > INSERT INTO Batting (MatchID,PID,POrder,HOut,FOW,NRuns,Mts,Nballs,Fours,Six)
--> VALUES (2755,23001,3,'C',51,0,12,6,0,0);

Query OK, 1 row affected (0.0040 sec)

MySQL localhost:33060+ ssl aarya61_match SQL > INSERT INTO Batting (MatchID,PID,POrder,HOut,FOW,NRuns,Mts,Nballs,Fours,Six)
--> VALUES (2755,25001,5,'C',232,71,104,80,4,0);

Query OK, 1 row affected (0.0028 sec)

MySQL localhost:33060+ ssl aarya61_match SQL > INSERT INTO Batting (MatchID,PID,POrder,HOut,FOW,NRuns,Mts,Nballs,Fours,Six)
--> VALUES (2755,91001,1,'C',74,60,85,52,8,2);

Query OK, 1 row affected (0.0022 sec)

MySQL localhost:33060+ ssl aarya61_match SQL > INSERT INTO Batting VALUES
--> (2755,94002,7,'LBW',157,17,23,29,1,0),
--> (2755,92002,11,'NO','NO',1,11,7,0,0),
--> (2689,89001,2,'C',205,91,176,121,7,0),
--> (2689,23001,4,'C',175,38,27,38,2,2),
--> (2689,25001,5,'C',240,36,52,37,2,1),
--> (2689,99002,1,'C',2,2,3,3,0,0),
--> (2689,95001,3,'C',8,1,9,7,0,0),
--> (2689,24001,5,'C',123,42,81,56,2,1),
--> (2689,99001,8,'B',228,7,20,12,0,0),
--> (2689,27001,9,'C',255,7,7,7,1,0),
--> (2755,27001,9,'B',257,2,7,6,0,0),
--> (2689,98002,8,'LBW',249,3,7,3,0,0),
--> (2755,98002,8,'RO',253,2,7,4,0,0);

Query OK, 13 rows affected (0.0046 sec)

```

6. DESCRIBE Batting;

```

MySQL localhost:33060+ ssl aarya61_match SQL > DESCRIBE Batting;
+-----+-----+-----+-----+-----+-----+
| Field | Type           | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| MatchID | int unsigned   | YES  | MUL | NULL    |       |
| PID      | int unsigned   | YES  | MUL | NULL    |       |
| POrder   | tinyint unsigned | YES  |     | NULL    |       |
| HOut     | varchar(5)     | YES  |     | NULL    |       |
| FOW      | varchar(5)     | YES  |     | NULL    |       |
| NRuns    | smallint unsigned | YES  |     | NULL    |       |
| Mts      | smallint unsigned | YES  |     | NULL    |       |
| Nballs   | tinyint unsigned | YES  |     | NULL    |       |
| Fours    | tinyint unsigned | YES  |     | NULL    |       |
| Six      | tinyint unsigned | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
10 rows in set (0.0034 sec)

```

- CREATE TABLE Bowling (

MatchID INT UNSIGNED,

PID INT UNSIGNED,

NOvers TINYINT UNSIGNED,

Maidens TINYINT UNSIGNED,

Nruns SMALLINT UNSIGNED,

Nwickets TINYINT UNSIGNED,

FOREIGN KEY (MatchID) REFERENCES ODIMatch(MatchID),

FOREIGN KEY (PID) REFERENCES Player(PlayerID));

INSERT INTO Bowling (MatchID,PID,NOvers,Maidens,Nruns,Nwickets)

VALUES (2689,99001,10,0,58,1);

INSERT INTO Bowling (MatchID,PID,Novers,Maidens,Nruns,Nwickets)

VALUES (2689,24001,3,0,27,1);

INSERT INTO Bowling VALUES

(2689,23001,3,0,15,0),

(2755,94002,9,1,40,1),

(2755,92002,10,0,56,1),

(2755,91001,4,0,29,0),

(2755,23001,10,0,53,2),

(2755,98002,10,0,40,3),

(2689,98002,10,0,44,1);

```
MySQL localhost:33060+ ssl aarya61_match SQL > CREATE TABLE Bowling (
-> MatchID INT UNSIGNED,
-> PID INT UNSIGNED,
-> Novers TINYINT UNSIGNED,
-> Maidens TINYINT UNSIGNED,
-> Nruns SMALLINT UNSIGNED,
-> Nwickets TINYINT UNSIGNED,
-> FOREIGN KEY (MatchID) REFERENCES ODIMatch(MatchID),
-> FOREIGN KEY (PID) REFERENCES Player(PlayerID));

Query OK, 0 rows affected (0.0380 sec)
MySQL localhost:33060+ ssl aarya61_match SQL >
MySQL localhost:33060+ ssl aarya61_match SQL > INSERT INTO Bowling (MatchID,PID,Novers,Maidens,Nruns,Nwickets)
-> VALUES (2689,99001,10,0,58,1);

Query OK, 1 row affected (0.0035 sec)
MySQL localhost:33060+ ssl aarya61_match SQL >
MySQL localhost:33060+ ssl aarya61_match SQL > INSERT INTO Bowling (MatchID,PID,Novers,Maidens,Nruns,Nwickets)
-> VALUES (2689,24001,3,0,27,1);

Query OK, 1 row affected (0.0025 sec)
MySQL localhost:33060+ ssl aarya61_match SQL >
MySQL localhost:33060+ ssl aarya61_match SQL > INSERT INTO Bowling VALUES
-> (2689,23001,3,0,15,0),
-> (2755,94002,9,1,40,1),
-> (2755,92002,10,0,56,1),
-> (2755,91001,4,0,29,0),
-> (2755,23001,10,0,53,2),
-> (2755,98002,10,0,40,3),
-> (2689,98002,10,0,44,1);

Query OK, 7 rows affected (0.0040 sec)

Records: 7 Duplicates: 0 Warnings: 0
```

8. DESCRIBE ODIMatch;

DESCRIBE Player;

DESCRIBE Bowling;

DESCRIBE Batting;

```
MySQL localhost:33060+ ssl aarya61_match SQL > DESCRIBE ODIMatch;
```

Field	Type	Null	Key	Default	Extra
MatchID	int unsigned	NO	PRI	NULL	
Team1	varchar(15)	YES		NULL	
Team2	varchar(15)	YES		NULL	
Ground	varchar(20)	YES		NULL	
Date	date	YES		NULL	
Winner	varchar(10)	YES		NULL	

6 rows in set (0.0031 sec)

```
MySQL localhost:33060+ ssl aarya61_match SQL > DESCRIBE Player;
```

Field	Type	Null	Key	Default	Extra
PlayerID	int unsigned	NO	PRI	NULL	
LName	varchar(15)	YES		NULL	
FName	varchar(15)	YES		NULL	
Country	varchar(25)	YES		NULL	
Yborn	int	YES		NULL	
Bplace	varchar(30)	YES		NULL	
FTest	int	YES		NULL	

7 rows in set (0.0017 sec)

```
MySQL localhost:33060+ ssl aarya61_match SQL > DESCRIBE Bowling;
```

Field	Type	Null	Key	Default	Extra
MatchID	int unsigned	YES	MUL	NULL	
PID	int unsigned	YES	MUL	NULL	
NOvers	tinyint unsigned	YES		NULL	
Maidens	tinyint unsigned	YES		NULL	
Nruns	smallint unsigned	YES		NULL	
Nwickets	tinyint unsigned	YES		NULL	

6 rows in set (0.0015 sec)

```
MySQL localhost:33060+ ssl aarya61_match SQL > DESCRIBE Batting;
```

Field	Type	Null	Key	Default	Extra
MatchID	int unsigned	YES	MUL	NULL	
PID	int unsigned	YES	MUL	NULL	
POrder	tinyint unsigned	YES		NULL	
HOut	varchar(5)	YES		NULL	
FOW	varchar(5)	YES		NULL	
NRuns	smallint unsigned	YES		NULL	
Mts	smallint unsigned	YES		NULL	
Nballs	tinyint unsigned	YES		NULL	
Fours	tinyint unsigned	YES		NULL	
Six	tinyint unsigned	YES		NULL	

10 rows in set (0.0030 sec)

Questions

Q1a. Print ODIMatch table.

```
SELECT * FROM ODIMatch;
```

MySQL localhost:33060+ ssl aarya61_match SQL > SELECT * FROM ODIMatch;

MatchID	Team1	Team2	Ground	Date	Winner
2324	Pakistan	India	Peshwar	2006-02-06	Team1
2327	Pakistan	India	Rawalpindi	2006-02-11	Team2
2357	India	England	Delhi	2006-03-28	Team1
2377	West Indies	India	Kingston	2006-05-18	Team2
2404	Sri Lanka	India	Colombo	2006-08-16	Abandoned
2440	India	Australia	Mohali	2006-10-29	Team2
2449	South Africa	India	Cape Town	2006-11-26	Team1
2480	India	West Indies	Nagpur	2007-01-21	Team1
2493	India	west Indies	Vadodara	2007-01-31	Team1
2520	India	Sri Lanka	Rajkot	2007-02-11	Team2
2611	England	India	Southampton	2007-08-21	Team1
2632	India	Australia	Mumbai	2007-10-17	Team1
2643	India	Pakistan	Guwahati	2007-11-05	Team1
2675	Australia	India	Melbourne	2008-02-10	Team2
2681	India	Sri Lanka	Adelaide	2008-02-19	Team1
2688	Australia	India	Sydney	2008-03-02	Team2
2689	Australia	India	Brisbane	2008-03-04	Team2
2717	Pakistan	India	Karachi	2008-06-26	Team2
2750	Sri Lanka	India	Colombo	2008-08-24	Team2
2755	Sri Lanka	India	Colombo	2008-08-24	Team2

20 rows in set (0.0031 sec)

Q1b. Print Player table.

```
SELECT * FROM Player;
```

MySQL localhost:33060+ ssl aarya61_match SQL > SELECT * FROM Player;

PlayerID	LName	FName	Country	Yborn	Bplace	FTest
21001	Sehwag	Virender	India	1978	Delhi	2001
23001	Singh	Yuvraj	India	1981	Chandigarh	2003
24001	Symonds	Andrew	Australia	1975	Birmingham	2004
25001	Dhoni	MS	India	1981	Ranchi	2005
27001	Kumar	Praveen	India	1986	Meerut	NULL
27002	Sharma	Ishant	India	1988	Delhi	2007
89001	Tendulkar	Sachin	India	1973	Mumbai	1989
90001	Lara	Brain	West Indies	1969	Santa Cruz	1990
90002	Kumble	Anil	India	1970	Bangalore	1990
91001	Jayasuriya	Sanath	Sri Lanka	1969	Matara	1991
92001	Warne	Shane	Australia	1969	Melbourne	1992
92002	Murlitharan	Muthiah	Sri Lanka	1972	Kandy	1992
92003	UI-Huq	Inzamam	Pakistan	1970	Multan	1992
93002	Streak	Heath	Zimbabwe	1974	Bulawayo	1993
93003	Kirsten	Gary	South Africa	1967	Cape Town	1993
94002	Vaas	Chaminda	Sri Lanka	1974	Mattumagala	1994
94004	Fleming	Stephen	New Zealand	1973	Christchurch	1994
95001	Ponting	Ricky	Australia	1974	Launceston	1995
95002	Pollock	Shaun	South Africa	1973	Port Elizabeth	1995
95003	Kallis	Jacques	South Africa	1975	Cape Town	1995
96001	Dravid	Rahul	India	1973	Indore	1996
96002	Gibbs	Herschelle	South Africa	1974	Cape Town	1996
96003	Ganguly	Saurav	India	1972	Calcutta	1996
97004	Vettori	Daniel	New Zealand	1979	Auckland	1997
98001	Afridi	Shahid	Pakistan	1980	Khyber Agency	1998
98002	Singh	Harbhajan	India	1980	Jalandhar	1998
99001	Lee	Breet	Australia	1976	Wollongong	1999
99002	Gilchrist	Adam	Australia	1971	Bellingen	1999
99003	Vaughan	Michael	England	1974	Manchester	1999

29 rows in set (0.0026 sec)

Q1c. Print Bowling table.

SELECT * FROM Bowling;

```
MySQL localhost:33060+ ssl aarya61_match SQL> SELECT * FROM Bowling;
```

MatchID	PID	NOvers	Maidens	Nruns	Nwickets
2689	99001	10	0	58	1
2689	24001	3	0	27	1
2689	23001	3	0	15	0
2755	94002	9	1	40	1
2755	92002	10	0	56	1
2755	91001	4	0	29	0
2755	23001	10	0	53	2
2755	98002	10	0	40	3
2689	98002	10	0	44	1

9 rows in set (0.0017 sec)

Q1d. Print Batting table.

SELECT * FROM Batting;

```
MySQL localhost:33060+ ssl aarya61_match SQL> SELECT * FROM Batting;
```

MatchID	PID	POrder	HOut	FOW	NRuns	Mts	Nballs	Fours	Six
2755	23001	3	C	51	0	12	6	0	0
2755	25001	5	C	232	71	104	80	4	0
2755	91001	1	C	74	60	85	52	8	2
2755	94002	7	LBW	157	17	23	29	1	0
2755	92002	11	NO	NO	1	11	7	0	0
2689	89001	2	C	205	91	176	121	7	0
2689	23001	4	C	175	38	27	38	2	2
2689	25001	5	C	240	36	52	37	2	1
2689	99002	1	C	2	2	3	3	0	0
2689	95001	3	C	8	1	9	7	0	0
2689	24001	5	C	123	42	81	56	2	1
2689	99001	8	B	228	7	20	12	0	0
2689	27001	9	C	255	7	7	7	1	0
2755	27001	9	B	257	2	7	6	0	0
2689	98002	8	LBW	249	3	7	3	0	0
2755	98002	8	RO	253	2	7	4	0	0

16 rows in set (0.0018 sec)

Q2. Find the Ids, First name, Last name as Surname of all cricket players.

SELECT PlayerID,Fname,Lname AS Surname FROM Player;

```
MySQL localhost:33060+ ssl aarya61_match SQL> SELECT PlayerID,Fname,Lname AS Surname FROM Player;
```

PlayerID	Fname	Surname
21001	Virender	Sehwag
23001	Yuvraj	Singh
24001	Andrew	Symonds
25001	MS	Dhoni
27001	Praveen	Kumar
27002	Ishant	Sharma
89001	Sachin	Tendulkar
90001	Brain	Lara
90002	Anil	Kumble
91001	Sanath	Jayasuriya
92001	Shane	Warne
92002	Muthiah	Murthitharan
92003	Inzamam	UI-Huq
93002	Heath	Streak
93003	Gary	Kirsten
94002	Chaminda	Vaas
94004	Stephen	Fleming
95001	Ricky	Ponting
95002	Shaun	Pollock
95003	Jacques	Kallis
96001	Rahul	Dravidi
96002	Herschelle	Gibbs
96003	Saurav	Ganguly
97004	Daniel	Vettori
98001	Shahid	Afridi
98002	Harbhajan	Singh
99001	Breet	Lee
99002	Adam	Gilchrist
99003	Michael	Vaughan

29 rows in set (0.0009 sec)

Q3a. Find all the information from table Player about players from India.

SELECT * FROM Player WHERE Country='India';

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT * FROM Player WHERE Country='India';
```

PlayerID	LName	FName	Country	Yborn	Bplace	FTest
21001	Sehwag	Virender	India	1978	Delhi	2001
23001	Singh	Yuvraj	India	1981	Chandigarh	2003
25001	Dhoni	MS	India	1981	Ranchi	2005
27001	Kumar	Praveen	India	1986	Meerut	NULL
27002	Sharma	Ishant	India	1988	Delhi	2007
89001	Tendulkar	Sachin	India	1973	Mumbai	1989
90002	Kumble	Anil	India	1970	Bangalore	1990
96001	Dravid	Rahul	India	1973	Indore	1996
96003	Ganguly	Saurav	India	1972	Calcutta	1996
98002	Singh	Harbhajan	India	1980	Jalandhar	1998

10 rows in set (0.0018 sec)

Q3b. Find the player Ids, First name & birth place of Indian cricket players.

SELECT PlayerID,Fname,Bplace FROM Player WHERE Country='India';

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT PlayerID,Fname,Bplace FROM Player WHERE Country='India';
```

PlayerID	Fname	Bplace
21001	Virender	Delhi
23001	Yuvraj	Chandigarh
25001	MS	Ranchi
27001	Praveen	Meerut
27002	Ishant	Delhi
89001	Sachin	Mumbai
90002	Anil	Bangalore
96001	Rahul	Indore
96003	Saurav	Calcutta
98002	Harbhajan	Jalandhar

10 rows in set (0.0016 sec)

Q4. Find all the information from table Player about players from India who were born after 1980

SELECT * FROM Player WHERE Country='India' AND Yborn > 1980;

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT * FROM Player WHERE Country='India' AND Yborn > 1980;
```

PlayerID	LName	FName	Country	Yborn	Bplace	FTest
23001	Singh	Yuvraj	India	1981	Chandigarh	2003
25001	Dhoni	MS	India	1981	Ranchi	2005
27001	Kumar	Praveen	India	1986	Meerut	NULL
27002	Sharma	Ishant	India	1988	Delhi	2007

4 rows in set (0.0014 sec)

Q5. Find the details of all matches played by Pakistan

SELECT * FROM ODIMatch WHERE Team1='Pakistan' OR Team2='Pakistan';

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT * FROM ODIMatch WHERE Team1='Pakistan' OR Team2='Pakistan';
```

MatchID	Team1	Team2	Ground	Date	Winner
2324	Pakistan	India	Peshwar	2006-02-06	Team1
2327	Pakistan	India	Rawalpindi	2006-02-11	Team2
2643	India	Pakistan	Guwahati	2007-11-05	Team1
2717	Pakistan	India	Karachi	2008-06-26	Team2

4 rows in set (0.0012 sec)

Q6a. Find the details of all matches played by India as a Second team.

SELECT * FROM ODIMatch WHERE Team2='India';

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT * FROM ODIMatch WHERE Team2='India';
```

MatchID	Team1	Team2	Ground	Date	Winner
2324	Pakistan	India	Peshwar	2006-02-06	Team1
2327	Pakistan	India	Rawalpindi	2006-02-11	Team2
2377	West Indies	India	Kingston	2006-05-18	Team2
2404	Sri Lanka	India	Colombo	2006-08-16	Abandoned
2409	South Africa	India	Cape Town	2006-11-26	Team1
2611	England	India	Southampton	2007-08-21	Team1
2675	Australia	India	Melbourne	2008-02-10	Team2
2688	Australia	India	Sydney	2008-03-02	Team2
2689	Australia	India	Brisbane	2008-03-04	Team2
2717	Pakistan	India	Karachi	2008-06-26	Team2
2750	Sri Lanka	India	Colombo	2008-08-24	Team2
2755	Sri Lanka	India	Colombo	2008-08-24	Team2

12 rows in set (0.0019 sec)

Q6b. Find the details of all matches played by India or Australia as Team1.

SELECT * FROM ODIMatch WHERE Team1='Australia' OR Team1='India';

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT * FROM ODIMatch WHERE Team1='Australia' OR Team1='India';
```

MatchID	Team1	Team2	Ground	Date	Winner
2357	India	England	Delhi	2006-03-28	Team1
2440	India	Australia	Mohali	2006-10-29	Team2
2480	India	West Indies	Nagpur	2007-01-21	Team1
2493	India	West Indies	Vadodara	2007-01-31	Team1
2520	India	Sri Lanka	Rajkot	2007-02-11	Team2
2632	India	Australia	Mumbai	2007-10-17	Team1
2643	India	Pakistan	Guwahati	2007-11-05	Team1
2675	Australia	India	Melbourne	2008-02-10	Team2
2681	India	Sri Lanka	Adelaide	2008-02-19	Team1
2688	Australia	India	Sydney	2008-03-02	Team2
2689	Australia	India	Brisbane	2008-03-04	Team2

11 rows in set (0.0012 sec)

Q6c. Find the match details Team2, Winner & ground in which India won the toss

SELECT Team2, Winner, Ground FROM ODIMatch WHERE Team1 = 'India';

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT Team2, Winner, Ground FROM ODIMatch WHERE Team1 = 'India';
```

Team2	Winner	Ground
England	Team1	Delhi
Australia	Team2	Mohali
West Indies	Team1	Nagpur
West Indies	Team1	Vadodara
Sri Lanka	Team2	Rajkot
Australia	Team1	Mumbai
Pakistan	Team1	Guwahati
Sri Lanka	Team1	Adelaide

8 rows in set (0.0015 sec)

Q7. Find Ids of all players that have bowled in an ODI match in the database

SELECT DISTINCT PID FROM Bowling;

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT DISTINCT PID FROM Bowling;
```

PID
23001
24001
91001
92002
94002
98002
99001

7 rows in set (0.0049 sec)

Q8. Find names of teams and grounds where India has played an ODI match outside India.

SELECT DISTINCT Team1,Ground FROM ODIMatch WHERE Team2='India';

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT DISTINCT Team1,Ground FROM ODIMatch WHERE Team2='India';
```

Team1	Ground
Pakistan	Peshwar
Pakistan	Rawalpindi
West Indies	Kingston
Sri Lanka	Colombo
South Africa	Cape Town
England	Southampton
Australia	Melbourne
Australia	Sydney
Australia	Brisbane
Pakistan	Karachi

10 rows in set (0.0013 sec)

Q9. Find Ids of all players that batted in match 2755

SELECT PID FROM Batting WHERE MatchID=2755;

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT PID FROM Batting WHERE MatchID=2755;
```

PID
23001
25001
91001
94002
92002
27001
98002

7 rows in set (0.0016 sec)

Q10a. Display list of ground names where Australia has played as Team1

SELECT Ground FROM ODIMatch WHERE Team1='Australia';

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT Ground FROM ODIMatch WHERE Team1='Australia';
```

Ground
Melbourne
Sydney
Brisbane

3 rows in set (0.0013 sec)

Q10b. Display sorted list of ground names where Australia has played as Team1

SELECT Ground FROM ODIMatch WHERE Team1='Australia' ORDER BY
Ground;

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT Ground FROM ODIMatch WHERE Team1='Australia' ORDER BY Ground;
```

Ground
Brisbane
Melbourne
Sydney

3 rows in set (0.0011 sec)

Q11. Find name of all players whose last name starts with Sin

SELECT Fname,Lname FROM Player WHERE Lname LIKE 'Sin%';

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT Fname,Lname FROM Player WHERE Lname LIKE 'Sin%';
```

Fname	Lname
Yuvraj	Singh
Harbhajan	Singh

2 rows in set (0.0011 sec)

Q12. Find Ids, First name of all players who played in match 2755.

```
SELECT PlayerID,Fname FROM Player WHERE PlayerID IN (SELECT PID
FROM Batting WHERE MatchID=2755);
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT PlayerID,Fname FROM Player WHERE PlayerID IN (SELECT
PID FROM Batting WHERE MatchID=2755);
```

PlayerID	Fname
23001	Yuvraj
25001	MS
91001	Sanath
94002	Chaminda
92002	Muthiah
27001	Praveen
98002	Harbhajan

7 rows in set (0.0052 sec)

Q13. Find match Ids of all matches in which Tendulkar batted.

```
SELECT MatchID FROM Batting WHERE PID IN (SELECT PlayerID FROM
Player WHERE Lname='Tendulkar');
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT MatchID FROM Batting WHERE PID
IN (SELECT PlayerID FROM Player WHERE Lname='Tendulkar');
```

MatchID
2689

1 row in set (0.0018 sec)

Q14. Find match information of all matches in which dhoni has batted.

```
SELECT MatchID,Team1,Team2,Ground,Date FROM ODIMatch WHERE MatchID
IN (SELECT MatchID FROM Batting WHERE PID IN(SELECT PlayerID FROM
Player WHERE Lname='Dhoni'));
```

```
MySQL localhost:33060+ ssl aarya61_match SQL > SELECT MatchID,Team1,Team2,Ground,Date FROM ODIMatch WHERE MatchID
IN (SELECT MatchID FROM Batting WHERE PID IN(SELECT PlayerID FROM Player WHERE Lname='Dhoni'));
```

MatchID	Team1	Team2	Ground	Date
2755	Sri Lanka	India	Colombo	2008-08-24
2689	Australia	India	Brisbane	2008-03-04

2 rows in set (0.0018 sec)